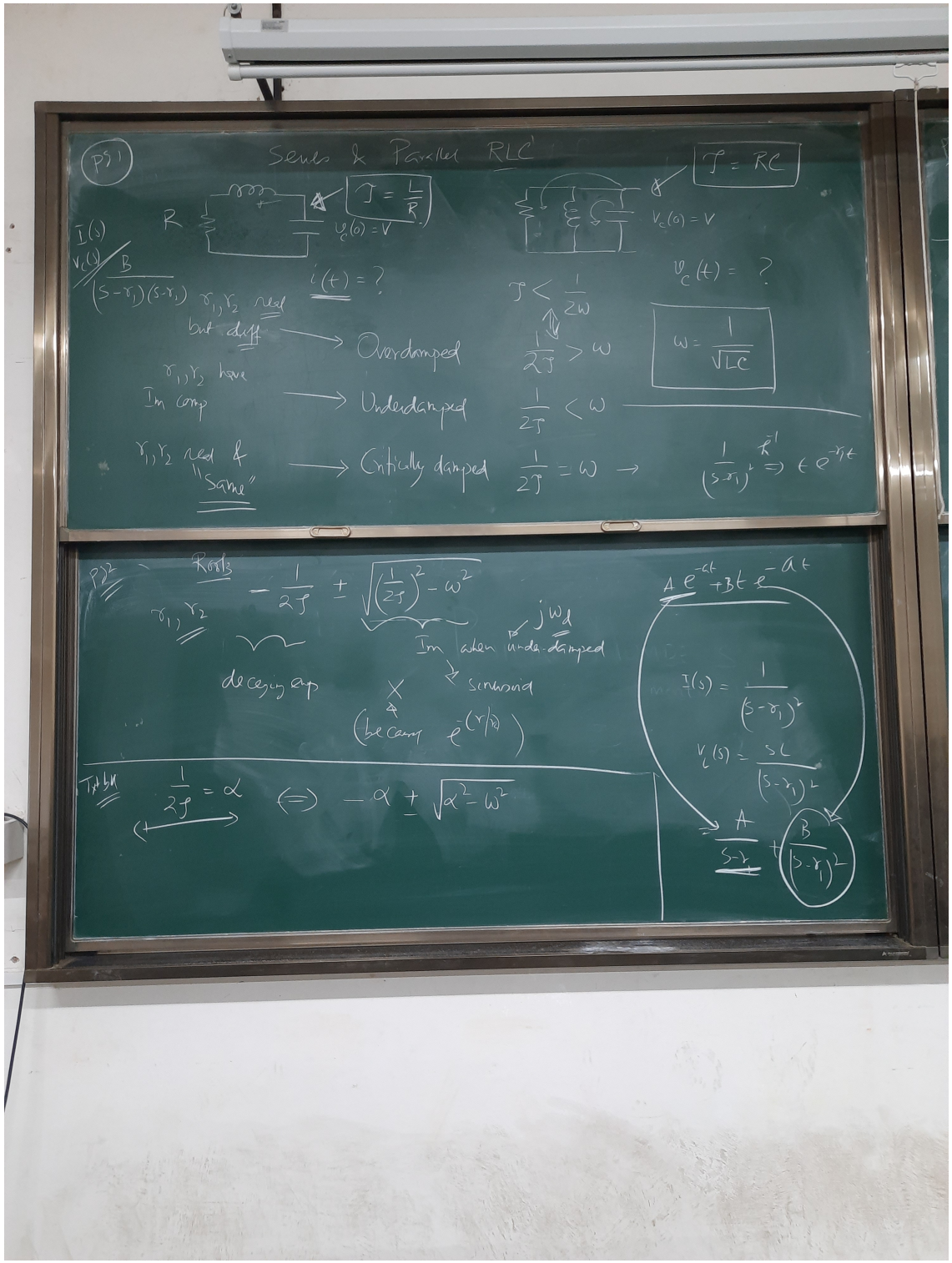
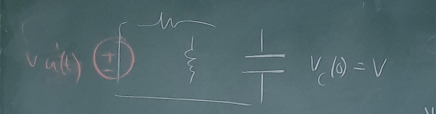

Series & Parallel RLC
τ = L/R
τ = RC
R
$v_c(0) = V$
$V_c(0) = V$
$I(s)$
$V_c(s)$
$\frac{B}{(s-r_1)(s-r_2)}$
$i(t) = ?$
$v_c(t) = ?$
r_1, r_2 real but diff → Overdamped
$\tau < \frac{1}{2\omega}$
$\frac{1}{2\tau} > \omega$
$\omega = \frac{1}{\sqrt{LC}}$
r_1, r_2 have Im comp → Underdamped
$\frac{1}{2\tau} < \omega$
r_1, r_2 real & "same" → Critically damped
$\frac{1}{2\tau} = \omega \rightarrow$
$\frac{1}{(s-r_1)^2} \overset{\mathcal{L}^{-1}}{\Rightarrow} t e^{-r_1 t}$
Roots
$-\frac{1}{2\tau} \pm \sqrt{\left(\frac{1}{2\tau}\right)^2 - \omega^2}$
r_1, r_2
$j\omega_d$
Im when under-damped
decaying exp
sinusoid
(because $e^{-(r/2)}$)
$Ae^{-at} + Bte^{-at}$
$I(s) = \frac{1}{(s-r_1)^2}$
$V_L(s) = \frac{sL}{(s-r_1)^2}$
$\frac{A}{s-r} + \frac{B}{(s-r_1)^2}$
Tut 6M
$\frac{1}{2\tau} = \alpha \iff -\alpha \pm \sqrt{\alpha^2 - \omega^2}$

PS3 RLC s/c source ⇒ transient response

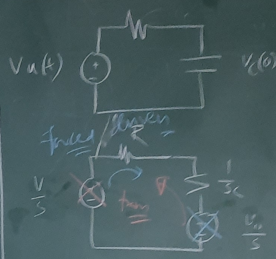


Both driven & transients??

↓

"Use superposition theorem"

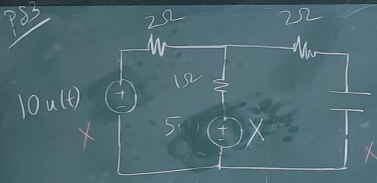
Treat capacitor charge as "source"



Use superposition

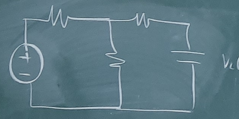
Total = trans + forced/driven

PS3



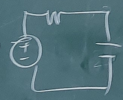
Use superposition

Case I



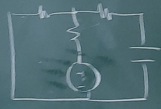
Use

Same trans



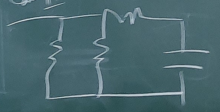


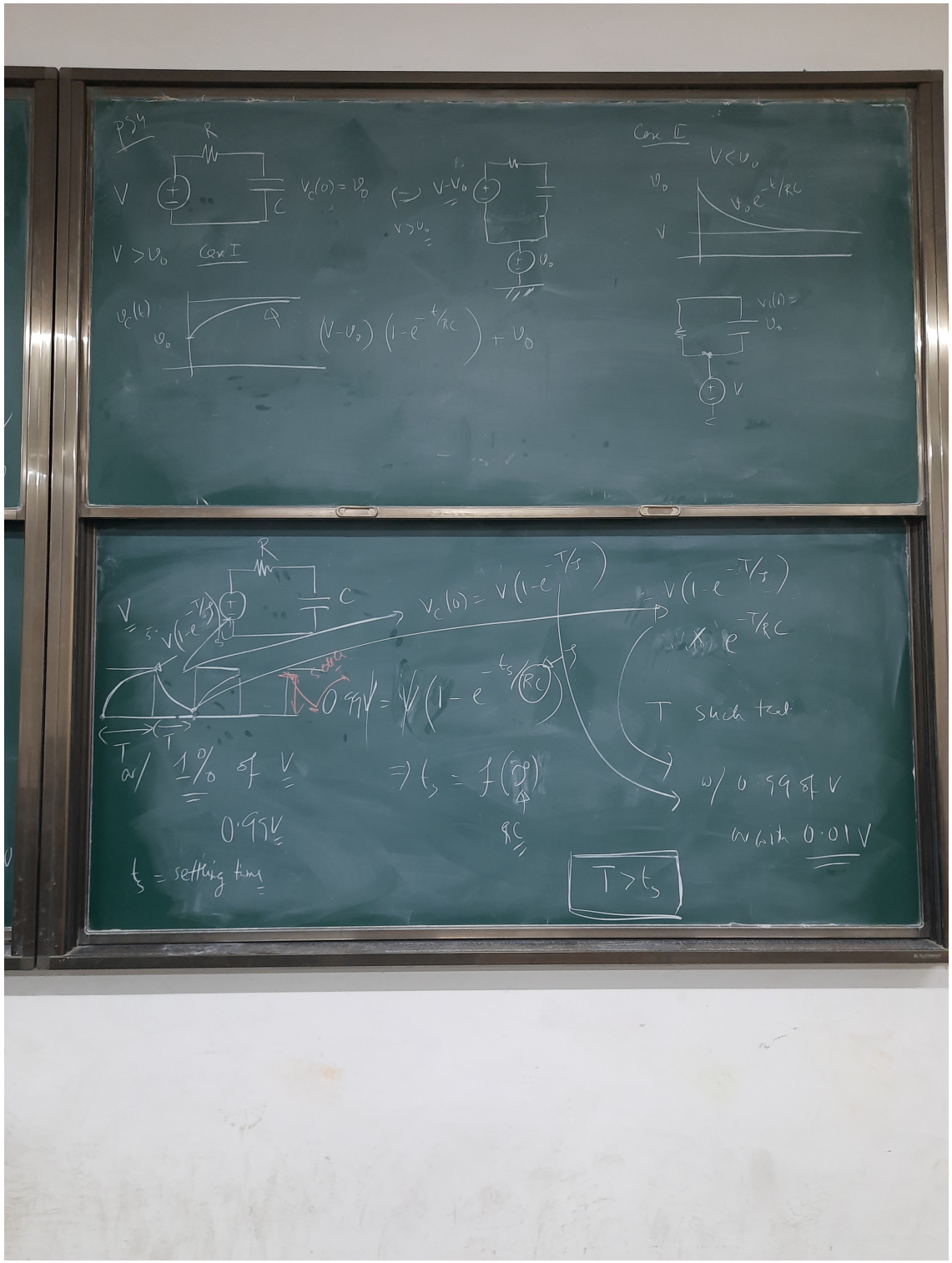
Case II



Same trans

Case II




PSU
R
V
C
$V_c(0) = V_0$
$V > V_0$
Case I
$V_c(t)$
V_0
$(V - V_0)\left(1 - e^{-t/RC}\right) + V_0$
$V - V_0$
$V > V_0$
V_0
Case II
$V < V_0$
V_0
$V_0 e^{-t/RC}$
V
$V_c(0) = V_0$
V
R
C
$V(1 - e^{-T/T_s})$
$V_c(0) = V(1 - e^{-T/T_s})$
$V(1 - e^{-T/T_s})$
$\times e^{-T/RC}$
$0.99V = V(1 - e^{-t_s/RC})$
T such that
w/ 0.99 of V
within 0.01 V
T
T_s
w/ 1% of V
0.99V
t_s = setting time
$\Rightarrow t_s = f(RC)$
$T > t_s$